INCH-POUND

MIL-R-55182/9E
26 January 1990
SUPERSEDING
MIL-R-55182/9D
15 July 1983

MILITARY SPECIFICATION

RESISTORS, FIXED, FILM, ESTABLISHED RELIABILITY STYLE RNC90 1/

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RNC90 1/, established reliability, fixed, film, resistors.

1.2 Classification.

1.2.1 Type designation. The type designation shall be in the following form:

RNC90	Y	562R00	B	M
Style and terminal type (1.2.1.1)	Characteristic (1.2.1.2)	Resistance (1.2.1.3)	Resistance tolerance (1.2.1.4)	Life failure rate (1.2.1.5)

1.2.1.1 Style and terminal type. Style and terminal type is identified by a three-letter symbol as indicated in table I and a two digit number. The letters identify established reliability, fixed, film, resistors of a specified terminal type and the number "90" identifies the size and configuration of the resistor applicable to this specification.

TABLE I. Style and terminal type.

Symbol	Terminal
RNC	Solderable/weldable (type C31, C32, or C51 of MIL-STD-1276)
RNN 1/	Weldable (type N-2 of MIL-STD-1276)

1/ Type N terminals are not available for resistance values less than 500 ohms (see 6.3).

1/ Third letter is variable, dependent upon lead material or capability.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Army Laboratory Command, ATTN: SLCET-R-S, Fort Monmouth, New Jersey 07703-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5905

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1.2.1.2 Characteristic. Style RNC90 is available in characteristic Y only. Characteristic Y is identified as follows:

Y - Nonhermetically sealed - Resistance temperature characteristic (referenced to +25°C) ± ±5 ppm/°C up to and including +125°C; and ±10 ppm/°C from +125°C to +175°C.

1.2.1.3 Resistance. The nominal resistance expressed in ohms is identified by six characters, consisting of five digits and a letter. The letter is used simultaneously as a decimal point and as a multiplier. For resistance values:

a. Greater than or equal to 1 ohm but less than 1,000 ohms, the letter "R" is used to represent a decimal point.

b. Greater than or equal to 1,000 ohms but less than 1 megohm, the letter "K" is used to represent a decimal point.

c. Greater than or equal to 1 megohm, the letter "M" is used to represent a decimal point.

All digits preceding and following the letter (R, K, or M) of the group represent significant figures. The resistance value designations are shown in table II. Minimum and maximum resistance values shall be as specified herein. The standard values for every decade shall follow the sequence specified in table IV of MIL-R-55182 for resistance tolerances D and F. The resistance values for tolerances Y, T, A, and B may be any value within the limits specified in 3.3, but it is preferred that values be chosen from the "B" and "D" column of table IV of MIL-R-55182.

TABLE II. Designation of resistance values.

Designation	Resistance
1R0000 to 9R8800 incl	1 to 9.88 ohms incl
10R000 to 98R800 incl	10 to 98.8 ohms incl
100R00 to 988R00 incl	100 to 988 ohms incl
1K0000 to 9K8800 incl	1 to 9.88 kilohms incl
10K000 to 98K800 incl	10 to 98.8 kilohms incl
100K00 to 988K00 incl	100 to 988 kilohms incl
1M0000 to 9M8800 incl	1 to 9.88 megohms incl

1.2.1.4 Resistance tolerance. The resistance tolerance is identified by a single letter in accordance with table III.

TABLE III. Resistance tolerance.

Symbol	Resistance tolerance
	Percent
Y	±0.005
T	±0.01
A	±0.05
B	±0.1
D	±0.5
F	±1.0

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1.2.1.5 Life failure-rate designation. The life failure-rate applicable to this specification shall be symbols M, P, R, and S.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-R-55182 - Resistors, Fixed, Film, Established Reliability, General Specification for.

STANDARD

MILITARY

MIL-STD-1276 - Leads for Electronic Component Parts.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-55182, and as specified herein.

3.2 Design and construction. Resistors shall be of the design, construction, and physical dimensions specified on figure 1.

3.3 Resistance. Resistance limits shall be as follows:

Minimum - 4.99 ohms
* Maximum - 200 kilohms

3.4 Power rating. The power rating shall be 0.30 watt (see 6.4).

3.5 Voltage rating. The maximum continuous working voltage shall not exceed 300 volts.

3.6 Thermal shock. When resistors are tested as specified, there shall be no evidence of mechanical damage; the change in resistance shall not exceed ±.05 percent for thermal shock and overload tests combined.

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3.7 Overload. When resistors are tested as specified with overload of 6.25 times rated power for 5 seconds, there shall be no evidence of arcing, burning, or charring; the change in resistance shall not exceed ±.05 percent for thermal shock and overload tests combined (see 3.6). The maximum voltage shall not exceed 450 volts.

3.8 Low-temperature operation. When resistors are tested as specified, there shall be no evidence of mechanical damage. The change in resistance shall not exceed ±.05 percent.

3.9 Terminal strength. When resistors are tested as specified in 4.2, there shall be no evidence of breaking or loosening of terminals from the resistors form, or chipping of coating, or other evidence of mechanical damage. The change in resistance shall not exceed ±.02 percent.

3.10 Dielectric withstanding voltage. When resistors are tested as specified in 4.3, there shall be no evidence of flashover, mechanical damage, arcing, or insulation breakdown. The change in resistance shall not exceed ±.02 percent.

3.11 Resistance to soldering heat. When resistors are tested as specified, there shall be no evidence of mechanical damage. The change in resistance shall not exceed ±.02 percent.

3.12 Moisture resistance. When resistors are tested as specified in 4.4, there shall be no evidence of mechanical damage. The change in resistance shall not exceed ±.05 percent. In addition, the dielectric withstanding voltage shall be as specified in 3.10, and the insulation resistance shall be 100 megohms, minimum.

3.13 Shock (specified pulse). When resistors are tested as specified, there shall be no evidence of mechanical or electrical damage. The change in resistance shall not exceed ±.01 percent. There shall be no electrical discontinuity during the test.

3.14 Vibration, high frequency. When resistors are tested as specified, there shall be no evidence of mechanical damage. The change in resistance shall not exceed ±.02 percent. There shall be no electrical discontinuity during the test.

3.15 Life.

3.15.1 Qualification. When resistors are tested as specified, there shall be no evidence of mechanical damage to the resistance element or enclosure. The change in resistance between initial measurement and any of the succeeding measurements up to and including 2,000 hours shall not exceed ±.05 percent.

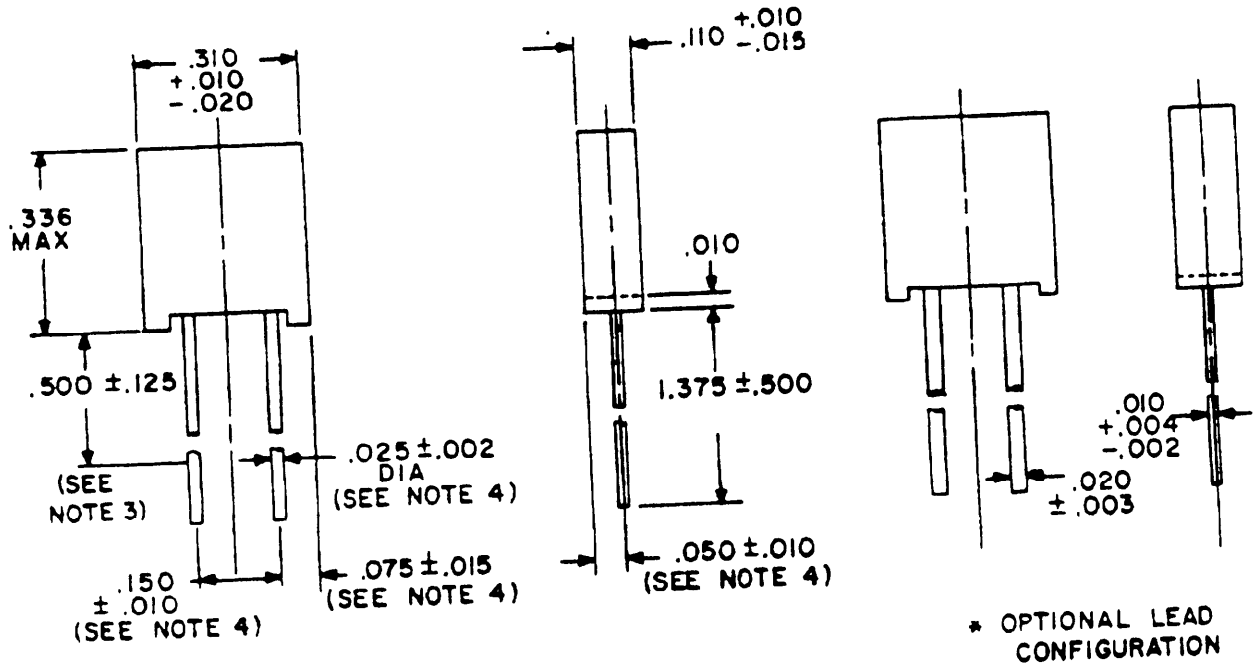
3.15.2 Failure-rate level determination. When resistors are tested as specified, there shall be no evidence of mechanical damage to the resistance element or enclosure. The change in resistance between the initial measurements and any of the succeeding measurements shall not exceed ±.5 percent. This single failure criteria shall be applicable to all measurements during the life test for purposes of determining failure-rate level qualification and is applicable as a parallel requirement with 3.15.1 to the measurements made during the life test specified for qualification inspection.

3.15.3 +70°C power rating. When resistors are tested as specified, there shall be no evidence of mechanical damage and the change in resistance shall not exceed ±.05 percent.

3.16 High temperature exposure. When resistors are tested as specified, there shall be no evidence of mechanical damage and the change in resistance shall not exceed ±.05 percent. Following this test, the dielectric withstanding voltage shall be as specified in 3.10, and the insulation resistance shall be 1,000 megohms, minimum.

3.17 Low temperature storage. When resistors are tested as specified, there shall be no evidence of mechanical damage. The change in resistance shall not exceed ±.05 percent.

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Inches	mm
.002	0.05
.003	0.08
.004	0.10
.010	0.25
.015	0.38
.020	0.51
.025	0.63
.050	1.27
.075	1.90
.110	2.79
.125	3.17
.150	3.81
.310	7.87
.336	8.53
.500	12.70
1.375	34.92

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Resistance measurement point.
4. The lead measurements shall be made at the point of emergency from the body.
*5. Style of standoffs optional.

FIGURE 1. Style RNC90 resistor.

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3.18 Voltage coefficient (applicable to resistors of 1,000 ohms and above). The voltage coefficient shall not exceed ±.0005 percent per volt.

3.19 Maximum weight. The maximum weight shall not exceed .00221 pound (gram).

*3.20 Marking. The resistor shall be marked as follows:

12345 - Source code.
3125J - Date code and JAN marking.
RNC90Y - Style, terminal, and characteristic.
49R000 - Resistance.
BM - Tolerance and failure rate.

Type designation shall be marked on one side of units, with remainder of required marking on other side.

NOTE: Optional marking is allowed on top of the surface to facilitate automatic printing equipment. The following is an example of optional marking:

a. Top: 05347J - Source code and JAN marking.

b. Side: RNC90Y - Style, terminal, and characteristic.
10K000B - Resistance and tolerance.
R8737A - Failure rate, date code, and lot code.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-55182, except as noted herein.

4.2 Terminal strength. In accordance with MIL-R-55182 except:

a. The body shall be clamped and the lead pulled in a direction parallel and along the axis of lead.

b. Applied force - 2 pounds.

4.3 Dielectric withstanding voltage. The test voltages applicable to style RNC90 are as follows:

Atmospheric pressure - 300 volts rms.
Barometric pressure (reduced) - 200 volts rms.

The resistor body shall be held between metal straps (see figure 2).

4.4 Moisture resistance. In accordance with MIL-R-55182, except the resistor body shall be held between metal straps during polarization (see figure 2).

5. PACKAGING

5.1 Packaging requirements. Packaging shall be in accordance with MIL-R-55182.

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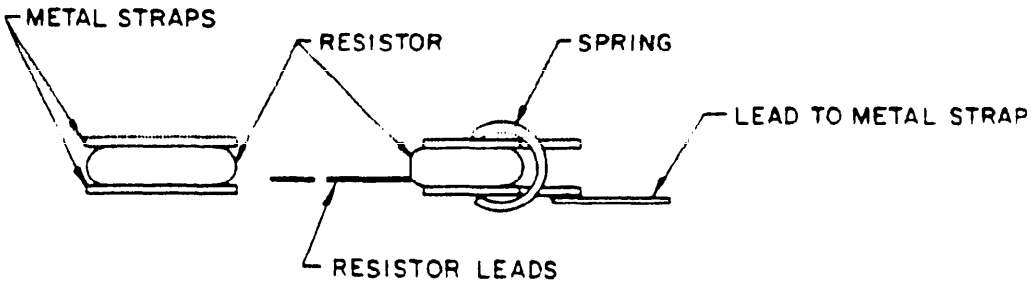


FIGURE 2. Metal strap assembly for dielectric withstanding voltage, moisture resistance and insulation resistance.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 The notes specified in MIL-R-55182 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

6.3 RNN resistors. Type RNN resistors are not available below 500 ohms value due to the temperature coefficient of nickle leads being of significant value when measurements of resistor TC of 5 ppm are made.

6.4 Power rating at +70°C. The power rating at +70°C for 2,000 hours of life test has been established at 0.6 watt, maximum voltage 300 volts. It should be noted that the failure-rate level is established at a single condition of +125°C, 0.3 watt, and 10,000-hour life test duration.

6.5 Supersession data. Prior to MIL-R-55182/9A, dated 20 March 1975, the four digit code for resistance specified in MIL-R-55182 applied.

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and the relationship to the last previous issue.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR
Navy - AS, OS
Air Force - 17, 99

User activities:
Army - AT, AV, ME
Navy - CG, MC

Preparing activity:
Army - ER

Agent:
DLA - ES

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